

SHAP Explainability for Phishing Email Detection

1. Introduction

Phishing email detection has progressed from manually constructed rules and conventional text classifiers towards deep neural architectures capable of modelling contextual language. However, a binary label and a confidence score do not establish why a system has classified a particular message as malicious or legitimate. In operational email screening, this distinction matters. Analysts must assess false positives, identify model reliance on misleading cues and communicate a defensible rationale to users. Explainable artificial intelligence (XAI) addresses this interpretability requirement, although an explanation should not be mistaken for proof of correctness or a causal account of an email's malicious intent.

This review critically evaluates SHapley Additive exPlanations (SHAP) as a post-hoc feature attribution approach for phishing email classification. It examines theoretical foundations, competing methods, empirical concerns and design implications for a project combining DeBERTa-v3-large contextual representations with twelve engineered email features. The emphasis here is the SHAP methodology itself; the precise software integration and validation of the hybrid architecture belong in a separate review and technical artefact.

2. Background and key concepts

2.1 Local explanation, model output and feature attribution

A local explanation describes a model's behaviour for one input rather than summarising performance across a population. For an email classifier, the target may be a class specific probability, logit or another model output. The choice of output scale is essential. A SHAP value is expressed in the units of the output being explained and is not essentially a probability. Accordingly, a user interface should not equate a numerical attribution with percentage confidence without an explicitly justified transformation (Lundberg and Lee 2017; SHAP developers n.d.-a).

2.2 Shapley values and additive decomposition

SHAP builds on cooperative game theoretic Shapley values. In an additive explanation, a baseline expectation and individual feature contributions reconstruct the explained output under the method's assumptions. Conceptually, $f(x) \approx E[f(X)] + \sum \phi_i$, where ϕ_i denotes the contribution attributed to feature i . The baseline, masking process, feature coalition rules and output link function determine the meaning of the contributions. Lundberg and Lee (2017) formalised a class of additive explanations and discussed desirable properties including local accuracy, missingness and consistency.

For a displayed phishing class explanation, a positive contribution pushes the explained output towards phishing and a negative contribution pushes it away, provided that the implementation explains that class in a consistent output space. If the interface instead explains the safe class output, the interpretation reverses. Ranking features by absolute $|\phi_i|$ identifies influential factors without indicating whether they support or oppose the displayed class. This distinction is central to the proposed "Factors Supporting", "Factors Working Against" and "Strongest SHAP Factors" panels.

2.3 Textual versus engineered features

Transformer based email classifiers process tokenized subject/body text, whereas engineered features may describe message structure, links, addresses or linguistic indicators. These are not interchangeable explanation units. Token level SHAP describes the effect of perturbing token groups under a text masker. The engineered feature SHAP describes changes to specified structured inputs under a potentially different masking distribution. Combining both into one chart without clarifying the feature space can produce a misleading impression of comparability. SHAP's text masking interface and partition based explanation illustrate how feature grouping and masking choices influence interpretation (SHAP developers n.d.-b; n.d.-c).

3. Review of existing research

3.1 Foundational SHAP framework

Lundberg and Lee (2017) introduced SHAP as a unified feature attribution framework, connecting previously distinct explanation methods through an additive representation. Its major contribution is conceptual coherence and attributions are tied to an explicit explanatory game rather than informal attention visualisations. This provides a principled basis for showing opposing and supporting influences on a specific classifier output. However, the framework does not guarantee that a chosen background distribution, masking operator or approximation reflects realistic email variation. Those remain application level decisions.

3.2 LIME and local surrogate explanation

Ribeiro, Singh and Guestrin (2016) proposed Local Interpretable Model-agnostic Explanations (LIME), which fits an interpretable surrogate around a prediction using perturbed examples. Its flexibility makes it relevant to email text classification, especially when a complex classifier cannot be directly inspected. Nevertheless, the surrogate's explanation depends on neighbourhood construction, feature representation and perturbation sampling. SHAP provides a more explicit attribution framework, but it also depends on how absent features and their coalitions are operationalised. Therefore, the methods should be compared empirically for stability, runtime and interpretability rather than treating SHAP as universally superior.

3.3 Integrated Gradients and gradient-based attribution

Sundararajan, Taly and Yan (2017) proposed Integrated Gradients (IG), motivated by sensitivity and implementation-invariance axioms. IG attributes a differentiable model output to inputs along a path from a baseline to the observed example. For transformer models, it offers a technically natural route to embedding-level or token-related attribution without the same number of model-agnostic perturbation queries. Yet the reference baseline, attribution target and aggregation from subword embeddings to human readable text remain consequential. IG and SHAP therefore address overlapping but non-identical questions; neither method should be treated as a ground-truth causal explanation.

3.4 Tree-specific SHAP and scope of generalisation

Lundberg et al. (2020) showed how efficient SHAP calculations for tree ensembles can support both local explanations and aggregated global insights. Their findings establish the value of systematic feature attribution in a particular model family, but Tree SHAP's computational benefits should not be transferred uncritically to a large transformer–feature-fusion architecture.

Explaining the complete gated-hybrid prediction may require a model-agnostic or architecture-compatible explainer and careful handling of correlated inputs.

3.5 Critiques of explanation faithfulness

Jain and Wallace (2019) demonstrated that neural attention weights are not necessarily faithful explanations of NLP predictions, cautioning against equating internal salience with decision rationale. Adebayo et al. (2018) showed that some visually plausible attribution maps can remain insensitive to model parameters or training labels, motivating randomisation-based sanity checks. Slack et al. (2020) further demonstrated that perturbation-based post-hoc explainers can be manipulated through adversarially constructed model behaviour. Together, these studies establish a critical constraint: attractive attribution graphics do not independently establish reliability, robustness or faithfulness.

4. Comparison of existing methods or solutions

Method	Primary mechanism	Strength for email detection	Principal limitation
SHAP	Shapley inspired additive attributions	Class specific supporting/opposing contributions; consistent explanatory vocabulary	Cost, background and masking dependence; approximation and correlation issues
LIME	Locally fitted interpretable surrogate	Intuitive local text features; model agnostic	Sensitivity to perturbation neighborhood and surrogate fidelity
Integrated Gradients	Path-integrated input gradients	Suited to differentiable transformers; efficient gradient access	Baseline/path choice; token aggregation and gradient saturation concerns
Attention visualization	Displays attention weights	Available from many transformer architectures	Attention weights do not necessarily explain output causally or faithfully
Rule/indicator display	Hand-authored observable cues	Easy to communicate and audit	Indicators are not necessarily what the trained model used

The comparison suggests a layered rather than competitive interpretation. Explicit email indicators help users understand observable message properties, while SHAP estimates how selected model inputs contribute to a specified output. The interface must distinguish these two evidence types. An attribution display is also distinct from classifier evaluation: accuracy, precision, recall, F1 and calibration assess prediction behaviour, whereas explanation faithfulness and stability require separate tests.

5. Research gaps and limitations

5.1 Dependence on background and masking assumptions

A Shapley-style attribution is conditional on the value function used to represent feature absence. Masking words can produce unnatural or semantically broken emails; replacing engineered features independently can create impossible combinations, such as an altered URL indicator without a corresponding change in text. Correlated inputs complicate the allocation of shared evidence. Hierarchical feature coalitions can partially address correlation structure, but their attribution semantics differ from unrestricted Shapley values (SHAP developers n.d.-c).

5.2 Stability, faithfulness and adversarial sensitivity

Small changes in subject wording, formatting or URL placement may alter local attributions even when the classification remains stable. Stability should therefore be measured across meaning-preserving perturbations and repeated explainer runs, where stochasticity is present. Fidelity checks should verify the explainer's additive reconstruction on the correct output scale and test whether removing highly attributed evidence affects model behaviour. However, deletion tests can themselves create out-of-distribution inputs, so no single faithfulness metric is definitive (Adebayo et al. 2018; Slack et al. 2020).

5.3 Computational constraints and communication risk

A large transformer can make repeated model evaluations expensive, particularly when SHAP is computed interactively. Latency, memory use and explanation reproducibility become practical deployment constraints. There is also a communication risk: end users may interpret a high SHAP value as proof that a phrase is inherently malicious, when it is instead a local attribution relative to a particular model and reference distribution. Likewise, high confidence does not establish correctness or calibration.

5.4 Evidence gap specific to the proposed implementation

The project's available description establishes an intended SHAP workflow and three explanation sections, but does not by itself establish the explainer algorithm, background dataset, masking policy, output units, approximation budget or empirical faithfulness. These should be documented from the actual source code and validated experimentally before making claims of accurate, stable or faithful explanations. The literature reviewed here supports the design rationale, not an unverified claim that the current prototype has resolved these limitations.

6. Implications for the proposed project

6.1 Explain the actual deployed prediction function

The most defensible design is to explain the exact class-specific output returned by the deployed inference path, including any preprocessing and feature-fusion behaviour that materially affects prediction. If the SHAP implementation explains only engineered features, its output must be labelled as a partial feature-level explanation rather than an explanation of the complete DeBERTa contextual pathway. If text and engineered features are both attributed, the chosen perturbation and aggregation mechanisms must be documented separately.

6.2 Design the explanation interface around direction and magnitude

The interface should state which class is explained and display the baseline/output scale.

“Factors Supporting the Prediction” should contain contributions aligned with that explained class;

“Factors Working Against the Prediction” should contain opposing contributions; and “Strongest SHAP Factors” should order factors by absolute magnitude while retaining signed values.

Explanatory copy should avoid statements such as “SHAP proves this email is phishing” or “this factor caused the attack”. More accurate language is “this factor increased the model’s phishing output relative to its baseline”.

6.3 Define a defensible evaluation protocol

A focused validation protocol should include:

- 1) numerical additivity checks for selected emails
- 2) comparison of attributions across safe and phishing examples
- 3) sensitivity testing under controlled, meaning preserving text changes
- 4) measurement of explanation latency and repeatability
- 5) qualitative inspection of whether highlighted factors are understandable to users

6.4 Connect explanation to security-oriented user experience

Streamlit and browser-extension interfaces should present the verdict, confidence and explanation in a coherent hierarchy. The phishing indicators should be explicitly labelled as observed email characteristics, not automatically presented as learned feature attributions. The explanation panel should remain available for both Safe and Phishing outcomes, including cases in which the strongest evidence opposes the predicted class. This supports critical scrutiny rather than selective presentation of reassuring evidence.

7. Conclusion

This provides a substantive rationale for using SHAP to complement complex phishing-email classification: its additive framework offers an intelligible representation of local supporting and opposing feature contributions (Lundberg and Lee 2017). Comparison with LIME, Integrated Gradients and attention-based visualisation shows that the choice of explanation method is shaped by model access, computational cost, baseline assumptions and the kind of question being asked. Crucially, prior work on sanity checks and adversarial manipulation demonstrates that explanations must be evaluated rather than accepted on visual plausibility alone.

For the proposed DeBERTa-v3-large gated-hybrid system, SHAP is therefore best understood as an auditable, model-dependent explanation layer, not a substitute for classification metrics, a causal account of phishing intent or proof of trustworthiness. The next technical review should examine how the explanation target, text processing and twelve engineered features are actually connected to the deployed gated-hybrid inference function, supported by implementation-level evidence.

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